Exercise 1

a) Since $x \neq 0$ when $x \to 0^+$ we just need to find out for what values of $\alpha \in \mathbb{R}$ $\lim_{x \to 0^+} \frac{x^\alpha}{x} = 0$

This is equivalent to $\lim_{x \to 0^+} x^{\alpha - 1} = 0$, which holds for $\alpha - 1 > 0$, i.e. $\alpha > 1$.

b) If we look at the seq. $x_n = n\pi$ we have that $x_n \to \infty$, $x_n > 0$ but $x_n^2 \sin^2(x_n) = 0$ hence x can't be $o(x^2 \sin^2(x))$ as $x \to +\infty$.

The converse is not true as well. This time we look at the sequence $x_n = \pi/2 + n\pi$

$\lim_{n \to +\infty} \frac{x_n^2 \sin^2(x_n)}{x_n} = \lim_{x \to \infty} (\pi/2 + n\pi) \cdot 1 = +\infty$, that contradict the fact that $x^2 \sin^2(x) = o(x)$ for $x \to +\infty$

c) We can pick $h = \sqrt{fg}$

Hence $\frac{f}{h} = \sqrt{\frac{g}{f}}$ and $\frac{h}{g} = \sqrt{\frac{g}{f}}$ as well.

Since $\lim_{x \to +\infty} \frac{f(x)}{g(x)} = 0$, using the continuity of square root we deduce that $\lim_{x \to \infty} \frac{f(x)}{h(x)} = \lim_{x \to \infty} \frac{h(x)}{g(x)} = 0$ as well, and therefore $f = o(h)$ and $h = o(g)$.

Exercise 2

a) <u>Base step</u> $(k=0)$.

Pick $p_0(x) = 1$ and $\alpha_0 = 0$.

$f(x)$ is continuous and differentiable since for $x \neq 0$ it is composition of continuous and differentiable functions.

<u>Inductive step</u> $(k \Rightarrow k+1)$

Suppose $f^{(k)}(x) = \frac{p_k(x)}{x^{\alpha_k}} e^{-\frac{1}{x^2}}$ and we know it to be continuous (i.e. f is C^k) and differentiable.

Then $f^{(k+1)}(x) = \left(\frac{p_k(x)}{x^{\alpha_k}} e^{-\frac{1}{x^2}}\right)' = \frac{p_k'(x)}{x^{\alpha_k}} e^{-\frac{1}{x^2}} + \frac{(-\alpha_k) p_k(x)}{x^{\alpha_k+1}} e^{-\frac{1}{x^2}} + \frac{2 p_k(x)}{x^{\alpha_k} x^3} e^{-\frac{1}{x^2}}$

$$= \frac{\left[x^3 p_k'(x) - \alpha_k x^2 p_k(x) + 2 p_k(x)\right]}{x^{\alpha_k+3}} e^{-1/x^2}$$

We can pick $\alpha_{k+1} = \alpha_k + 3$ and $p_{k+1}(x) = x^3 p_k'(x) - \alpha_k x^2 p_k(x) + 2 p_k(x)$ where p_{k+1} is a polynomial since the derivative p_k' is.

Then $f^{(k+1)}(x) = \frac{p_{k+1}(x)}{x^{\alpha_{k+1}}} e^{-1/x^2}$ and it is continuous and differentiable for $x \neq 0$ being composition and product of differentiable function. (i.e. f is $C^{k+1}(\mathbb{R})$).

b) <u>Base step</u> $(k=0)$

Defining $f(0) = 0$ f is continuous since $\lim_{x \to 0} e^{-1/x^2} = e^{-\infty} = 0$.

<u>Inductive step</u> $(k \Rightarrow k+1)$

~~Suppose defining $f^{(k)}(0) = 0$ $\forall$ $k \leq k$ makes f~~

Suppose $f^{(k)}(0) = 0$.

Let's prove that $f^{(k)}$ is differentiable at 0 and $f^{(k+1)}(0) = 0$.

$$f^{(k+1)}(0) = (f^{(k)})'(0) = \lim_{x \to 0} \frac{f^{(k)}(x) - f^{(k)}(0)}{x - 0}$$

$$= \lim_{x \to 0} \frac{f^{(k)}(x)}{x} = \lim_{x \to 0} \frac{p_k(x)}{x^{\alpha_{k+1}}} e^{-1/x^2} = 0$$

Since e^{-1/x^2} goes to 0 super exponentially $\Rightarrow$ the thesis is proved

c) Since $f^{(k)}(0) = 0$ $\forall$ $k \in \mathbb{N}$ the Taylor series of f is simply given by $\sum_{n=0}^{\infty} 0 = 0$ $\forall$ $x \in \mathbb{R}$.

This does not converge to $f(x)$ $\forall$ $x \neq 0$ since $e^{-1/x^2} > 0$ if $x \neq 0$.

Exercise 3

a) $\lim_{x\to 0} \frac{\sin x - x^2}{\sqrt{1+x} - 1} \overset{(\text{Hopital})}{=} \lim_{x\to 0} \frac{\cos(x) - 2x}{\frac{1}{2\sqrt{1+x}}} = \frac{1-0}{1/2} = 2$

$\lim_{x\to 0} \frac{\sin x - x^2}{\sqrt{1+x} - 1} = \lim_{x\to 0} \frac{x + o(x) + o(x)}{\cancel{1} + \frac{1}{2}x + o(x) \cancel{-1}} = \lim_{x\to 0} \frac{\cancel{x} + o(x)}{\frac{1}{2}\cancel{x} + o(x)} = \lim \frac{1 + o(1)}{\frac{1}{2} + o(1)} = 2$

b) $\lim_{x\to 0} \frac{\sin x - x + \frac{1}{6}x^3}{x^5} \overset{(H)}{=} \lim_{x\to 0} \frac{\cos x - 1 + \frac{1}{2}x^2}{5x^4} \overset{(H)}{=} \lim_{x\to 0} \frac{-\sin x + x}{20x^3} \overset{(H)}{=}$

$= \lim_{x\to 0} \frac{-\cos x + 1}{60x^2} \overset{(H)}{=} \lim_{x\to 0} \frac{\sin x}{120x} \overset{(H)}{=} \lim_{x\to 0} \frac{\cos(x)}{120} = \frac{1}{120}$

$\lim_{x\to 0} \frac{\sin x - x + \frac{1}{6}x^3}{x^5} = \lim_{x\to 0} \frac{\cancel{x} \cancel{-} \cancel{\frac{1}{6}x^3} + \frac{1}{120}x^5 \cancel{-x} + \cancel{\frac{1}{6}x^3} + o(x^5)}{x^5}$

$= \lim_{x\to 0} \frac{\frac{1}{120} + o(1)}{1} = \frac{1}{120}$

At any application of De l'Hopital thm (denoted by (H)) we checked that the limit of both the numerato at the denominator was 0.

Exercise 4

a) $\lim_{x\to 0} \frac{f(x)}{g(x)} = \lim_{x\to 0} \frac{\sqrt{1+x}}{1+x} = \frac{\sqrt{1}}{1} = 1$

$\lim_{x\to 0} \frac{f'(x)}{g'(x)} = \lim_{x\to 0} \frac{\frac{1}{2}\frac{1}{\sqrt{1+x}}}{1} = \frac{\frac{1}{2}\cdot\frac{1}{\sqrt{1}}}{1} = \frac{1}{2} \neq 1.$

The theorem does not apply since $\lim_{x\to 0} f(x) \neq 0$ and $\lim_{x\to 0} g(x) \neq 0$

b) $\lim_{x\to 0} f(x) = \lim_{x\to 0} x + \sin(1/x)x^2 = \lim_{x\to 0} x + o(x) = 0$ $\left(\begin{array}{l} |\sin(1/x)| \leq 1 \\ \Rightarrow \sin(1/x)x^2 = o(x) \end{array} \right)$

$\lim_{x\to 0} g(x) = \lim_{x\to 0} x = 0.$

$\lim_{x\to 0} \frac{f(x)}{g(x)} = \lim_{x\to 0} \frac{x + o(x)}{x} = \lim_{x\to 0} \frac{1 + o(1)}{1} = \frac{1}{1} = 1$

But $f'(x) = 1 + 2x\sin(1/x) - \cos(1/x)$, $g'(x) = 1$

$\frac{f'(x)}{g'(x)} = \underbrace{1 + 2x\sin(1/x)}_{A(x)} - \underbrace{\cos(1/x)}_{B(x)}$

$A(x)$ has 1 as a limit for $x\to 0$, but $B(x)$ has no limit. It follows that $\frac{f'(x)}{g'(x)} = A(x) + B(x)$ has no limit at 0.

Exercise 5

a) We fix $x \in \mathbb{R}$ and we need to check that
$\sum_{n=0}^{\infty} f_n(x) = f_\infty(x)$
This is trivial for $x=0$. For $x \neq 0$ it becomes

$$\sum_{n=0}^{\infty} \frac{x^2}{(1+x^2)^n} = x^2 \cdot \frac{1}{1-\frac{1}{1+x^2}}$$ since it is a geometric series and $1+x^2 > 1 \ \forall x \neq 0$

$$= x^2 \cdot \frac{1}{\frac{x^2}{1+x^2}} = 1+x^2 = f_\infty(x).$$

b) We just compute

$$\lim_{x\to 0} f_\infty(x) = \lim_{x \to 0} 1+x^2 = 1$$

$$\lim_{n\to\infty} \sum_{k=0}^{n} f_k(0) = \lim_{n\to\infty} \sum_{k=0}^{n} 0 = 0. \neq 1.$$

c) For $x \neq 0$ let's compute

$$f_\infty(x) - \sum_{k=0}^{n} f_n(x) = \sum_{k=n+1}^{\infty} \frac{x^2}{(1+x^2)^k}$$

$$= x^2 \cdot \frac{1}{(1+x^2)^{n+1}} \frac{1}{1-\frac{1}{x^2+1}}$$

$$= \frac{x^2+1}{(x^2+1)^{n+1}} = \frac{1}{(1+x^2)^n}$$

If $x \geq \frac{1}{2}$ then $1+x^2 \geq 5/4$

And $\left| f_\infty(x) - \sum_{k=0}^{n} f_k(x) \right| \leq \left(\frac{4}{5}\right)^n$

Fix $\varepsilon > 0$, let n_0 be such that $\left(\frac{4}{5}\right)^{n_0} < \varepsilon$.

Then $\left| f_\infty(x) - \sum_{k=0}^{n} f_k(x) \right| \leq \varepsilon \quad \forall \ n \geq n_0$ and $\forall \ x \in [1/2, 1]$.

$\Rightarrow \sum f_n$ converges uniformly on $[1/2, 1]$.

On the contrary suppose $\sum_{k=0}^{n} f_k(x)$ converges uniformly on $[0,1]$.

then $\forall \varepsilon > 0 \ \exists \ n_0$ s.t. $\left| f_\infty(x) - \sum_{k=0}^{n} f_k(x) \right| < \varepsilon \ \forall \ n \geq n_0 \ \forall \ x \in [0,1]$

In particular $\frac{1}{(1+x^2)^{n_0}} < \varepsilon \ \forall \ x \in (0,1)$.

But $\lim_{x\to 0} \frac{1}{(1+x^2)^{n_0}} = \frac{1}{1^{n_0}} = 1 > \varepsilon$ contradiction.